

EX.NO : 04

DATE :

ANALYZING THE PERFORMANCE OF VARIOUS CONFIGURATIONS AND PROTOCOLS IN LAN MULTI-ROUTING CONNECTION

AIM :

To establish connection between two LANs by extending multi-routing connection.

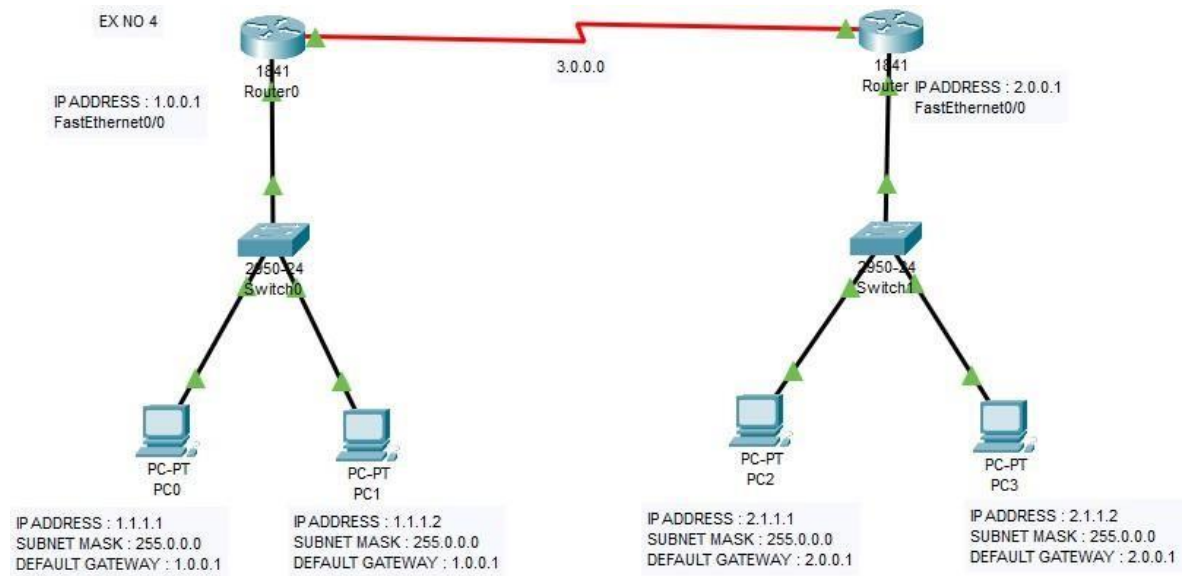
REQUIREMENTS :

- Four windows PC.
- Two Switch (2950-24).
- Six Straight Line LAN Cables.
- Two Router (1841).
- One Serial DTE Cable.
- Cisco Packet Tracer.

PROCEDURES :

- Open CISCO PACKET TRACER software.
- Draw the Four PC using END Device Icons.
- Draw the Cisco 24 Port Switch Using Switch icon lists.
- Draw the Cisco 1841 Routers using Router icon lists.
- Make the Connections using Copper-Straight-Through Ethernet Cables and Serial DTE Cables.
- Enter the IP Address To Each Machine.
- Configure Routers 0,1 and Serial 0/0 for 2Routers.
- Check the Network Connections using Add Simple PDU(P).

NETWORK TOPOLOGY :



HOST PC IP ADDRESS :

PC0 IP CONFIGURATION

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 1.1.1.1

Subnet Mask 255.0.0.0

Default Gateway 1.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

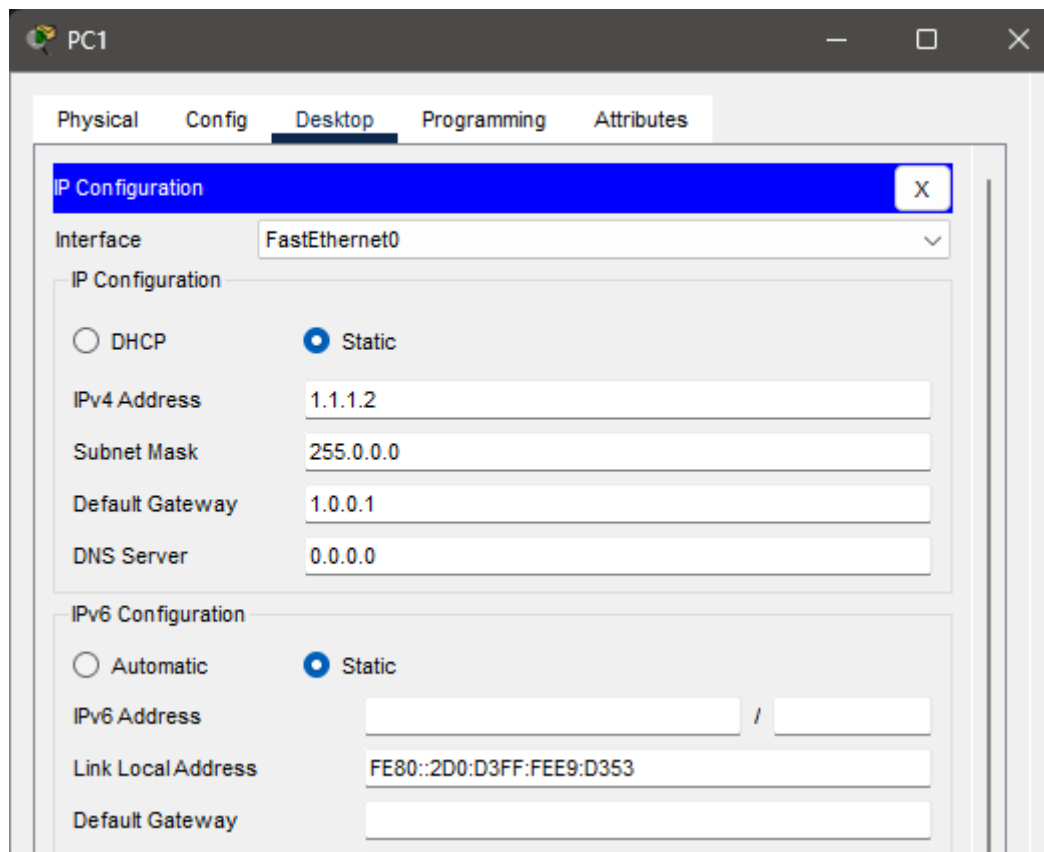
☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::201:97FF:FEEC:9E19

Default Gateway

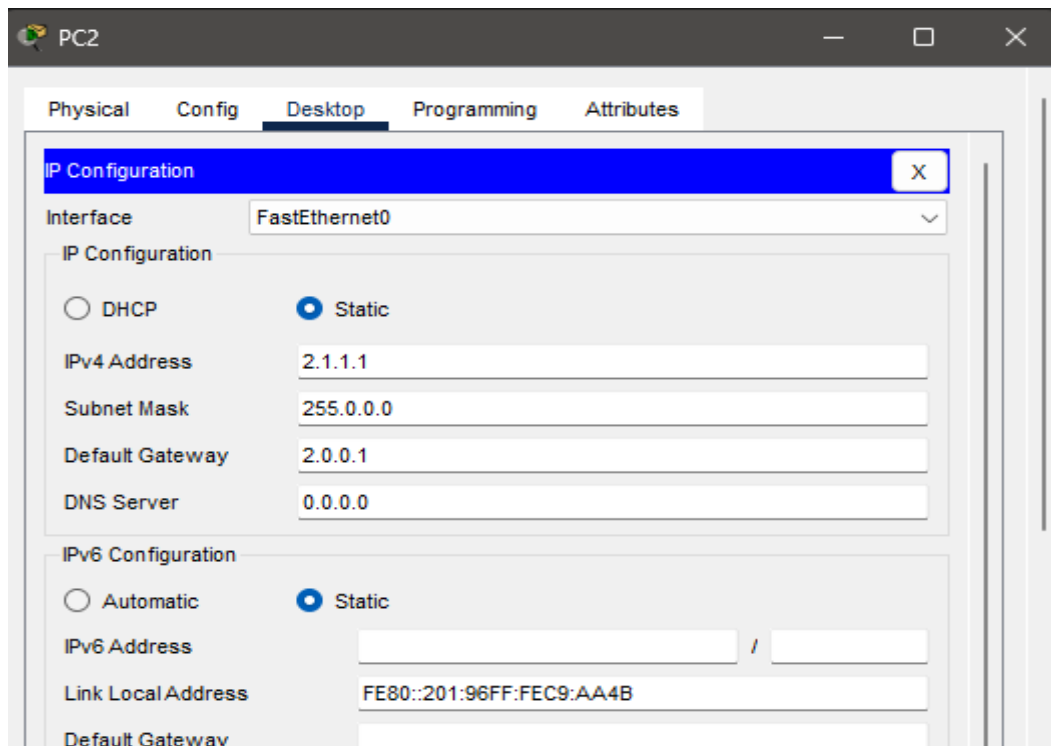
PC1 IP CONFIGURATION



The screenshot shows the 'PC1' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IPv4 Address: 1.1.1.2, Subnet Mask: 255.0.0.0, Default Gateway: 1.0.0.1, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The fields are: IPv6 Address (empty), Link Local Address: FE80::2D0:D3FF:FEE9:D353, and Default Gateway (empty).

Section	Field	Value
IP Configuration	Interface	FastEthernet0
	Configuration Type	Static
	IPv4 Address	1.1.1.2
	Subnet Mask	255.0.0.0
IPv6 Configuration	Configuration Type	Static
	IPv6 Address	
	Link Local Address	FE80::2D0:D3FF:FEE9:D353
	Default Gateway	

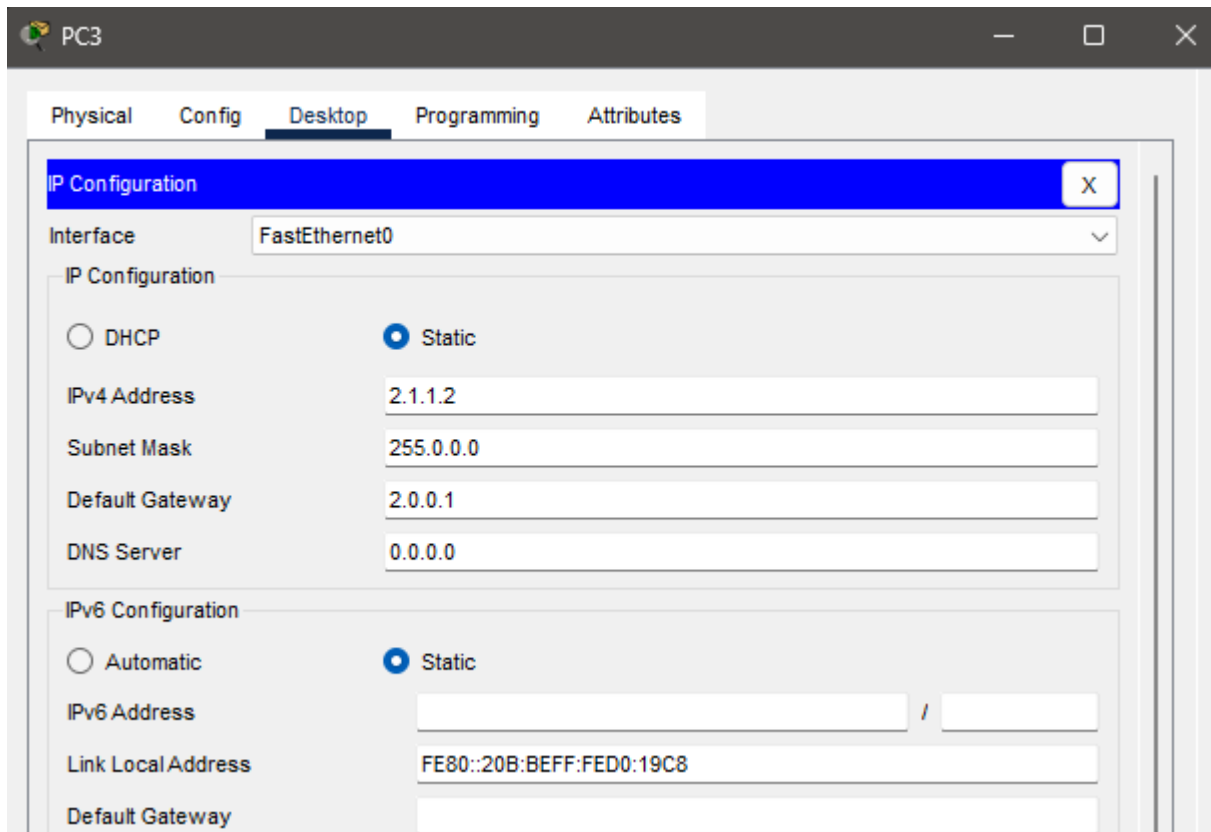
PC2 IP CONFIGURATION



The screenshot shows the 'PC2' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is highlighted in blue. The 'Interface' is set to 'FastEthernet0'. Under 'IP Configuration', the 'Static' radio button is selected. The fields are filled with: IPv4 Address: 2.1.1.1, Subnet Mask: 255.0.0.0, Default Gateway: 2.0.0.1, and DNS Server: 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is also selected. The fields are: IPv6 Address (empty), Link Local Address: FE80::201:96FF:FEC9:AA4B, and Default Gateway (empty).

Section	Field	Value
IP Configuration	Interface	FastEthernet0
	Configuration Type	Static
	IPv4 Address	2.1.1.1
	Subnet Mask	255.0.0.0
IPv6 Configuration	Configuration Type	Static
	IPv6 Address	
	Link Local Address	FE80::201:96FF:FEC9:AA4B
	Default Gateway	

PC3 IP CONFIGURATION



The image shows a window titled "PC3" with a dark header bar containing standard window controls. Below the header is a tabbed interface with four tabs: "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is selected and highlighted. Within the "Desktop" tab, there is a sub-tab titled "IP Configuration" with a close button "X" in its top right corner. The main content area of the "IP Configuration" sub-tab is divided into two sections. The top section is for "FastEthernet0" and contains two radio buttons: "DHCP" (unselected) and "Static" (selected). Below these are four text input fields: "IPv4 Address" with the value "2.1.1.2", "Subnet Mask" with "255.0.0.0", "Default Gateway" with "2.0.0.1", and "DNS Server" with "0.0.0.0". The bottom section is for "IPv6 Configuration" and also has two radio buttons: "Automatic" (unselected) and "Static" (selected). Below these are three text input fields: "IPv6 Address" (empty), "Link Local Address" with the value "FE80::20B:BEFF:FED0:19C8", and "Default Gateway" (empty).

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 2.1.1.2

Subnet Mask 255.0.0.0

Default Gateway 2.0.0.1

DNS Server 0.0.0.0

IPv6 Configuration

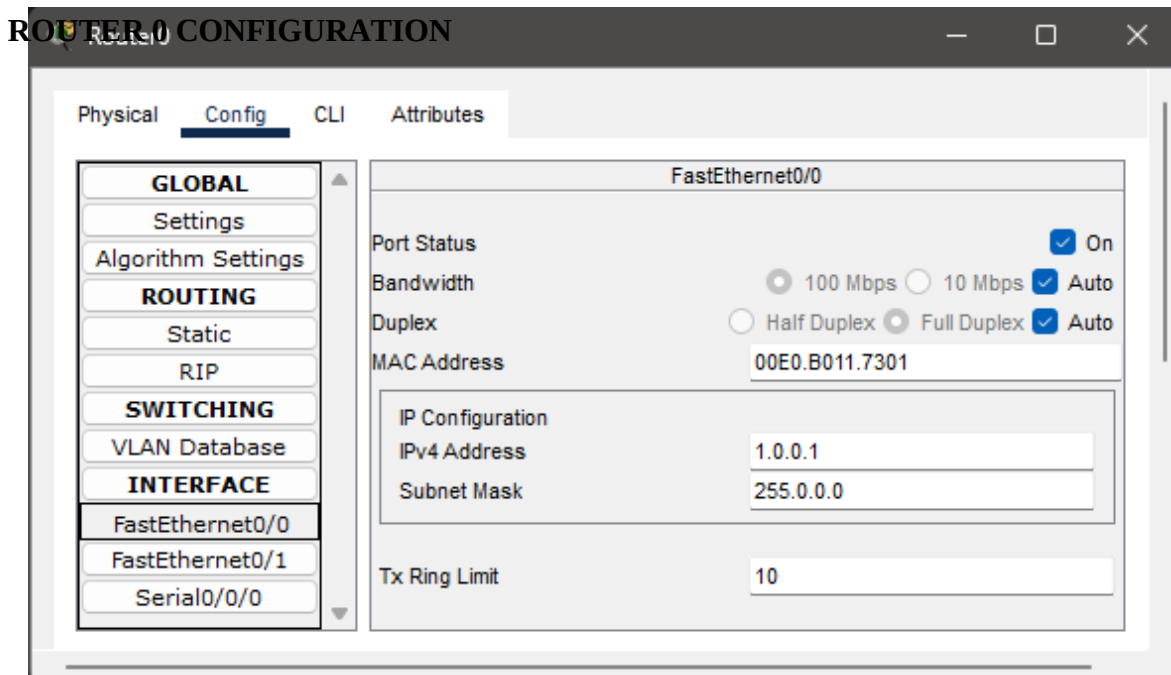
☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20B:BEFF:FED0:19C8

Default Gateway

ROUTERS CONFIGURATION:



The image shows a window titled "ROUTER" with a dark header bar containing standard window controls. Below the header is a tabbed interface with four tabs: "Physical", "Config", "CLI", and "Attributes". The "Config" tab is selected and highlighted. On the left side of the "Config" tab is a vertical sidebar with a list of configuration categories: "GLOBAL", "ROUTING", "SWITCHING", and "INTERFACE". Under "GLOBAL" are "Settings" and "Algorithm Settings". Under "ROUTING" are "Static" and "RIP". Under "SWITCHING" is "VLAN Database". Under "INTERFACE" are "FastEthernet0/0", "FastEthernet0/1", and "Serial0/0/0". The "FastEthernet0/0" item is selected and highlighted. The main content area of the "Config" tab is titled "FastEthernet0/0" and contains several configuration options. "Port Status" has a checked checkbox "On". "Bandwidth" has two radio buttons: "100 Mbps" (selected) and "10 Mbps" (unselected). "Duplex" has two radio buttons: "Half Duplex" (unselected) and "Full Duplex" (selected). "MAC Address" has a text input field with the value "00E0.B011.7301". Below these is a sub-section titled "IP Configuration" with two text input fields: "IPv4 Address" with the value "1.0.0.1" and "Subnet Mask" with the value "255.0.0.0". At the bottom is a "Tx Ring Limit" text input field with the value "10".

ROUTER

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

FastEthernet0/0

FastEthernet0/1

Serial0/0/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps

Duplex ☐ Half Duplex ☒ Full Duplex

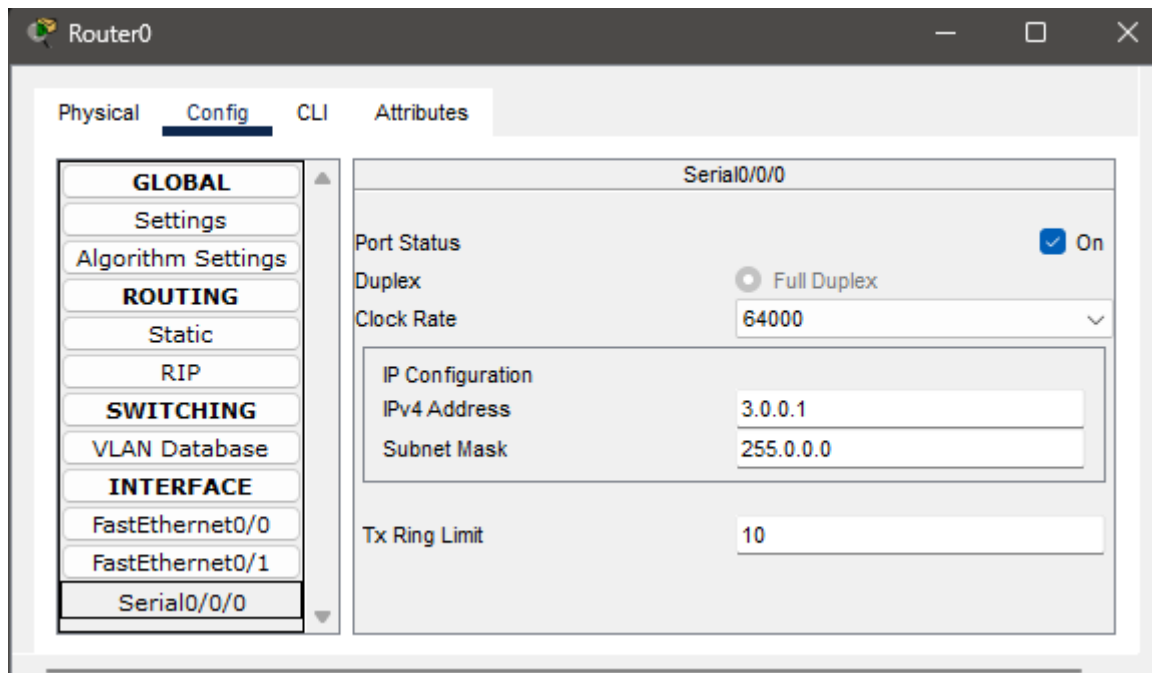
MAC Address 00E0.B011.7301

IP Configuration

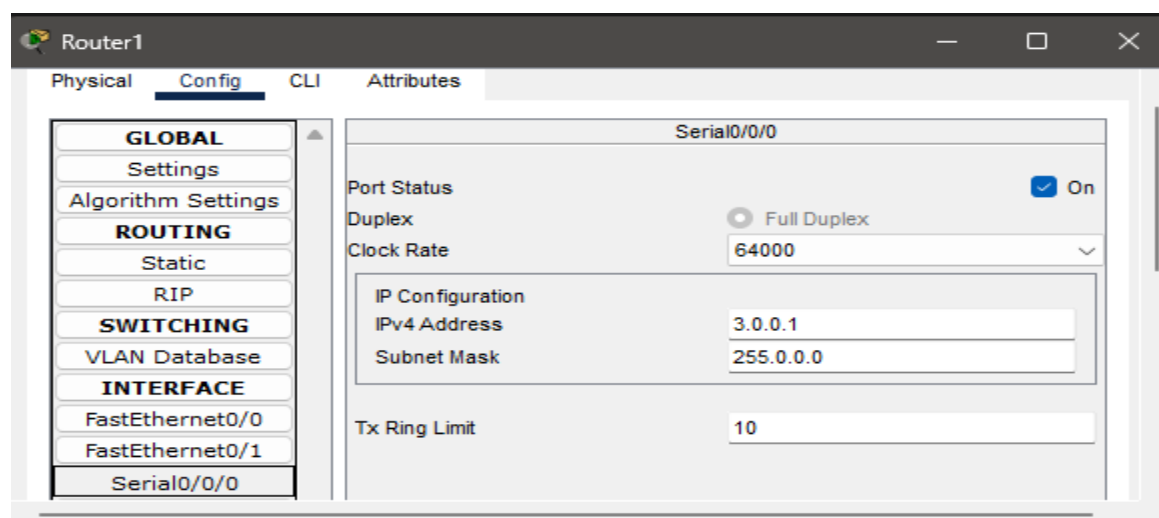
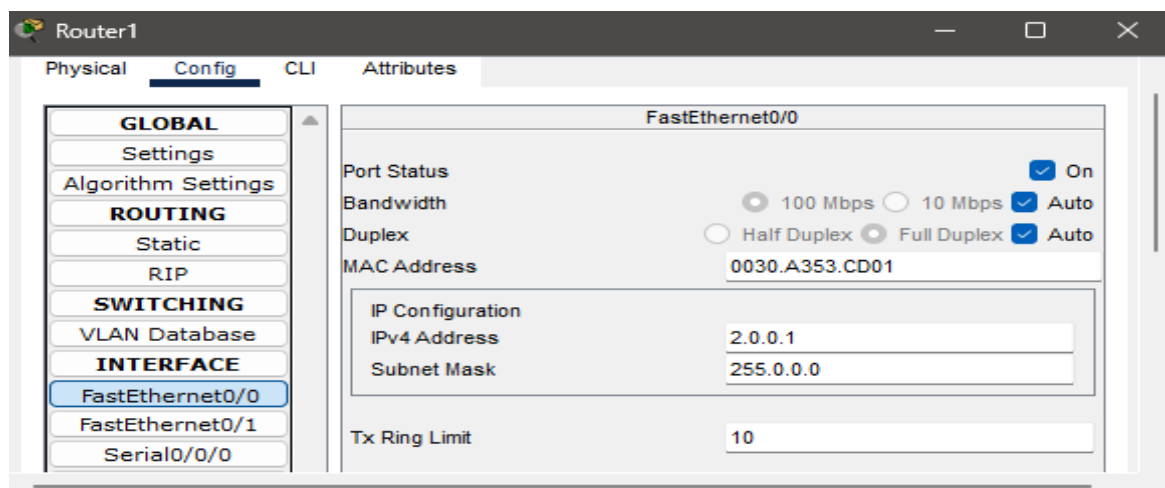
IPv4 Address 1.0.0.1

Subnet Mask 255.0.0.0

Tx Ring Limit 10

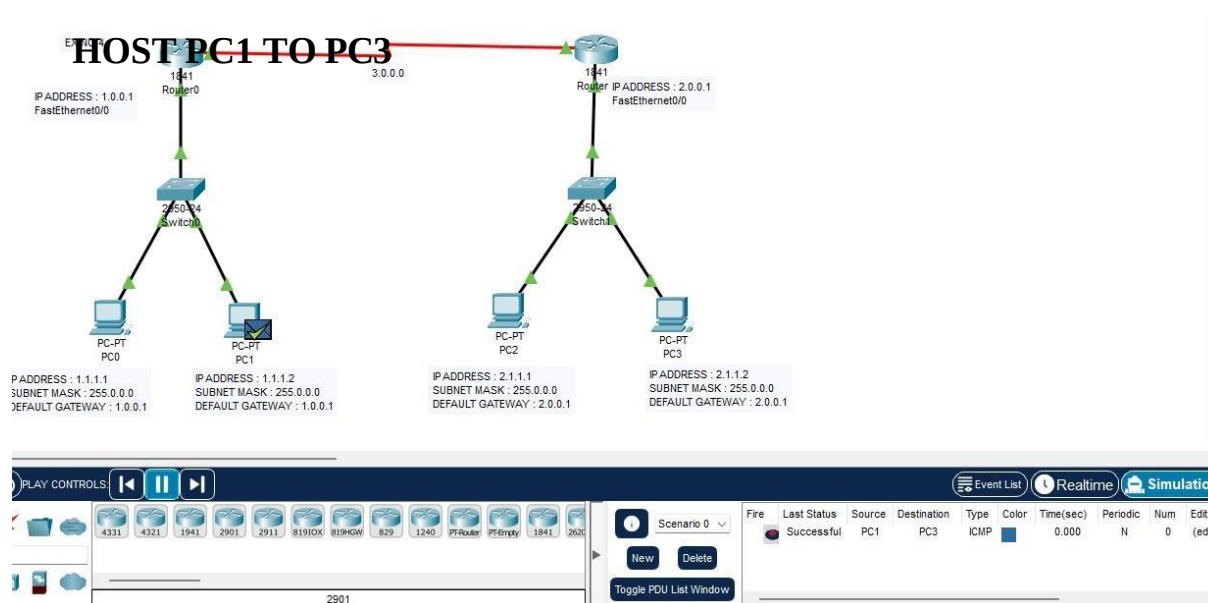
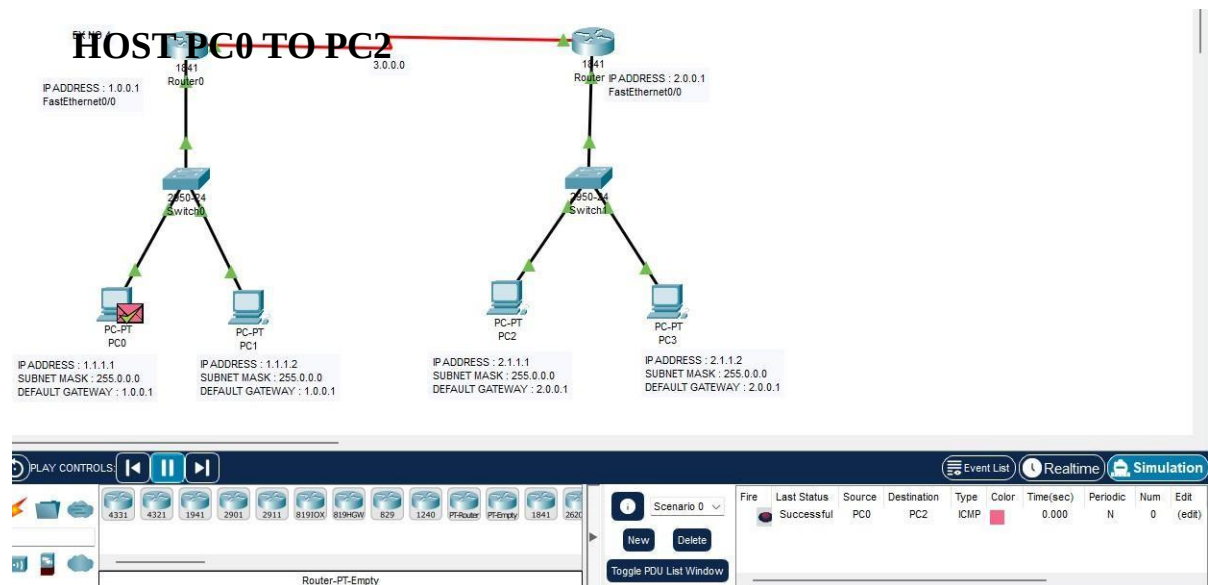


ROUTER 1 CONFIGURATION :



VERIFY LAN NETWORK CONNECTIVITY :

Using Add Simple PDU(p), Click the mail icon and then drop one mail to one of the PC in first lan and another mail to PC in another lan. If the resultant window shows the successful delivery, then network connectivity is successful.



RESULT :

Thus, two LANs are connected using multiple routers and the communication between LANs is checked successfully.

